Notes for chapter 4, sections 4-1, 4-2, & 4-3

Read pp. 81 – 82 (just the top half of p. 82), 85 (the second half of the page), & 89

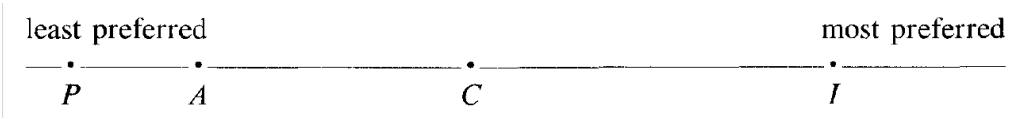
pp. 81 & 82

The utility values that you first encountered on pp. 24 – 25 were values on an ***ordinal scale***. An ordinal scale gives us the order of a person's perferences, but nothing else. For instance, here are four items and their utility values on an ordinal scale:

ice cream	3
cola	2
apples	1
popcorn	0

This is part of Resnik's example (on p. 82), and so, let's say that these are his utility values. Since these utility values are on an ordinal scale, we know, for instance, that he values ice cream more than cola (i.e., soda), but we don't know how much more he likes ice cream than soda.

Let's say that we ask him to put his relative preferences for these four items on a line where the distance between each item represents how much more or less he values the item relative to the others. He produces this (which is also on p. 82):



Now we know that, relative to popcorn and ice cream (which are the highest and lowest), apples are pretty close to popcorn and cola is about exactly in between popcorn and ice cream. So, if we want to more accurately represent Resnik's preferences, we can look at the line and estimate what his utility values will be on an **interval scale**. (Resnik doesn't exactly do this on the second half of p. 82, but don't worry about what he's doing there.)

ice cream	10
cola	5
apples	2
popcorn	0

Now we know that (for Resnik), ice cream is twice as valuable as cola and cola is more than twice as valuable as apples.

pp. 85

We will need a better method for determining utility values that are on an interval scale that just estimating they go on a number line. But before we get to that, let's think about why utility values that are on an interval scale are useful. Up until now, we have had two ways of representing the value of the outcomes in a decision problem: (1) utility values on an ordinal scale and (2) monetary values.

Utility values on an ordinal scale are useful because they let us represent the order of the decision maker's personal preferences. For instance, different decision makers might order four free trips in these ways:

Tokyo	4	Rocky Mtn National Park	4	Florida Keys	4
Paris	3	Paris	3	Tokyo	3
Rocky Mtn National Park	2	Florida Keys	2	Paris	2
Florida Keys	1	Tokyo	1	Rocky Mtn National Park	1

As has already been mentioned, the weakness of the ordinal scale is that it doesn't represent how valuable each trip is to the decision maker (relative to the other trips). Also, if we have a decision problem with risk, we can't use utility values on an ordinal scale to do an expected value-type calculation.

Alternatively, when we use monetary values, we are able to represent exactly how valuable each trip is relative to the others (and relative to everything else that has a monetary value). Here, for instance, are some reasonable current estimates of the monetary values of the four trips (based on the flights, hotels, exchange rates, and daily expenditures):

Florida Keys	$4,000
Paris	$3,500
Tokyo	$3,000
Rocky Mtn National Park	$2,500

As it works out, the value of each increases by $500 (and the trip to the Rockies is worth 62.5% of the trip to the Florida Keys, the trip to Tokyo is work 75% of the trip to the Florida Keys, etc.). But as you've probably noticed, ordering these trips by their monetary values doesn't reflect the order of any of our decision makers' preferences. (But even if it did, it still might not reflect how much they valued each trip relative to the others.)

Now, this is not to say that monetary values are not useful. And often they do reflect how much we value an items. (Or just how much we value the money itself.) This is what Resnik discusses on p. 85.

One could argue that Resnik is making a subtle mistake in the first paragraph on p. 85. He says that money "often works as an interval scale too—at least over a limited range of items." In contrast, I would say that money is always an interval scale. $10 is twice as much as $5. $100 is twice as much as $50. $1,000 is twice as much as $500. $10 million is twice as much as $5 million, and so on. But that said, for many people, $5 million is (or would be) so important that they value it almost as much as they value $10 million (or, at least, they place more value on it than half of $10 million). So, in this paragraph, Resnik fails to distinguish these two points: (1) money itself is an interval scale and (2) our subjective value of money sometimes deviates from that interval scale.

You can read p. 86 if you like. It's about using monetary values and the expected value strategy to make policy decisions. As Resnik explains in the last full paragraph on this page, despite its shortcomings, this is probably the best method for making such decisions.

But do read p. 87 starting with "On a personal level, too, the 'true value' of an alternative is often above or below its EMV."

There are different ways to set up the decision problem in Resnik's first example. He does it this way.

	P = .80 **investment is successful**	P = .20 **investment fails**
invest	$50,000	-$15,000
don't invest	$0	$0

If you do the calculation, you'll find that the expected value of investing is $37,000. If you don't invest, then you neither gain nor lose anything—and so you will still have your $15,000. So, following the expected value strategy, we should make the investment.

But what point is Resnik making with the example? Which is more valuable to you (the 'you' in the story), *investing* (which includes the possibility of losing the $15,000) or *not investing* (which guarantees that you'll keep your $15,000)?

They don't affect the point that he is making, but Resnik makes some mistakes in example 2. He describes it this way:

	P = 1/12 = .08 **roll two 1s**	P = 11/12 = .92 **roll anything else**
play	$20	-$10
don't play	$10	$10

(The probability of rolling two 1s is 1/36 = 2.7%. And the outcomes should be either what you end up with after the game is over: $20, $0, $10, and $10; or what you get from playing or not playing: $10, -$10, $0 and $0.)

But in any case, none of that effects which option has the higher expected value. The expected value is always going to be higher for *don't play* than for *play*.

What point is Resnik making with this example? Despite the expected values, what is the person in the example going to do?

The final point that Resnik discusses in this section concerns a general problem for the expected value strategy. (And this is a problem that isn't resolved by using utility values instead of monetary values.) Here is the starting point for decision problem:

	P = .000001 **your ticket is selected**	P = .999999 **your ticket isn't selected**
keep your ticket	$1,000,000	$0

If you calculate the expected value, you'll find that it is $1. But $1 doesn't seem to represent the experience of having the ticket, which will deliver either $1 million or $0.

pp. 88 - 89

In section 4-3, Resnik introduces a precise method for setting utility values on an interval scale. This may seem a little unusual at first. And it is a method that will be deployed differently for every person. So, Resnik (and I in my examples) won't be explaining exactly how it will go for you. That means that it's important to understand how and why the process works so that you can then use it for yourself.

In Resnik's example on p. 89, there are three outcomes that are going to be put on an interval utility scale (for the "you" in the example). They are, from most to least preferred,

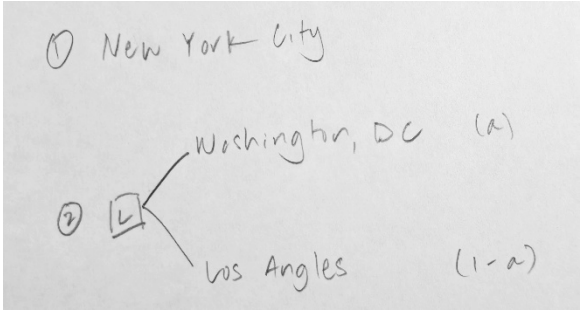
a trip to Washington, DC	1
a trip to New York City	__
a trip to Los Angeles	0

Because they are the most and least preferred, we begin by assigning a utility value of 1 to Washington, DC and 0 to Los Angeles. To set the utility value for New York City, we do what Resnik describes here (on p. 89):

York, but we can measure that by asking you the following question: Suppose you were offered a choice between a trip to New York and a lottery that will give you a trip to Washington if you "win" and one to Los Angeles if you "lose." How great a chance of winning would you need to have in order to be indifferent between these two choices? Presumably, if you prefer New York quite a bit

So, you (hypothetically) offer yourself two choices: (1) taking the trip to New York City or (2) entering a lottery. If you win the lottery, you get your most preferred outcome (Washington, DC). If you lose the lottery, you get your least preferred outcome (Los Angeles).

I often diagram it like this:



Your task, then, is to determine the probability of winning the lottery that would make you indifferent between (1) and (2). (In the diagram, I've represented the probability of winning and losing the lottery as a and $1 - a$. [This a has nothing to do with the optimism-pessimism rule, by the way.])

If you want a tangible way of thinking about the lottery, you can imagine either of the following.

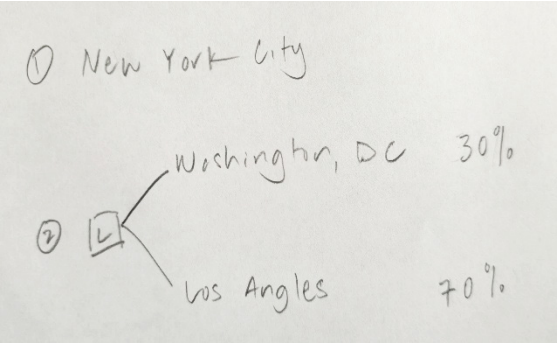
(1) In terms of the chance that a red marble will be randomly selected from a jar. If the jar contains 50 percent red marbles, then you'll have a 50 percent chance of winning the lottery (and a 50 percent chance of losing). If the jar contains 60 percent red marbles, then you'll have a 60 percent chance of winning the lottery (and a 40 percent chance of losing), and so on.

(2) In terms of the number of times that you'll win the lottery if you enter it 100 times. If the probability of winning the lottery is .50, then you'll win about 50 times and lose about 50 times. If the probability of winning the lottery is .60, then you'll win about 60 times and lose about 40 times. And so on.

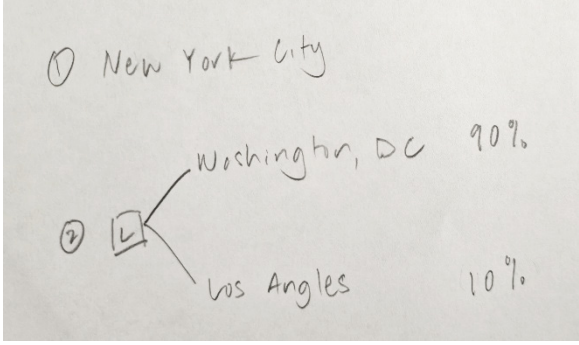
Being indifferent between choices (1) and (2) in this context means that you would be equally happy taking New York City or entering the lottery.

To begin thinking about how we determine a utility value with this method, imagine that we have two decision makers, Jeff and Mary, and for both of them WDC **P** NYC and NYC **P** LA. But that's where the similarity ends. Jeff is indifferent between getting New York City and the first lottery, and Mary is indifferent between getting New York City and the second lottery.

Jeff is indifferent between these two.



Mary is indifferent between these two.



So, Jeff doesn't care if he gets (1) New York City or (2) a lottery that he has a 30 percent chance winning and a 70 percent chance of losing.

Mary, meanwhile, doesn't care if she gets (1) New York City or (2) a lottery that she has a 90 percent chance of winning and a 10 percent chance of losing.

Which one of them values the trip to New York City more?

Mary does. Here are some points to keep in mind:

(*a*) The lottery is always between the most preferred and the least preferred items.

(*b*) Once we find the point where someone (or you) is indifferent between (1) and (2), we know that if the lottery gets safer, then the person will choose the lottery.

(*c*) And if the lottery gets riskier, then the person will take (1)—in this case, New York City.

Jeff is indifferent between NYC and a lottery that has a very high chance (a 70% chance) of delivering him his least preferred outcome, LA. So, getting NYC isn't too much more valuable to him than getting Los Angeles. And if the lottery got just a little bit safer—say, it was 35% and 65%—Jeff would take the lottery. He's willing to accept a small chance at getting his most preferred outcome (WDC), because it's much more valuable to him than NYC.

Mary, meanwhile, needs a very safe lottery before she becomes indifferent. If the lottery were any riskier—say, 85% and 15%—she would just take (1), New York City. For her, New York City is much more valuable than LA. She's willing to accept only a small chance of ending up with LA because NYC is almost as valuable for her as Washington, DC.

As Resnik explains, once we know where someone is indifferent between New York City and the lottery, we can add the lottery to the list of preferences. In Resnik's example, you are indifferent between New York and the lottery when there is a 75 percent chance of getting WDC (and a 25 percent chance of getting LA). So, this will be the list. (This is like the list on the bottom of p. 24.)

a trip to Washington, DC
a trip to New York City, L(.75, WDC, LA)
a trip to Los Angeles

If we assign 1 to Washington, DC and 0 to Los Angeles, then we know this lottery will get the value .75. Since you are indifferent between New York City and this lottery, then for you, New York also gets the value .75. (Read Resnik's explanation about this right after the list on p. 89.)

Thus, these are your utility value on an interval scale:

a trip to Washington, DC	1
a trip to New York City	.75
a trip to Los Angeles	0

Note how Resnik represents the lottery: L(a, x, y). x is always the most preferred item, and y is always the least preferred item. a is the probability of getting the most preferred item.

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the expected utility strategy

Once we have the utility values for each outcome, we can use the ***expected utility strategy***—which is similar to the expected value strategy—to make a decision. Let's add one more item so that we have four outcomes, and let this be the decision problem:

	P = .40 **red marble is drawn**	P = .60 **blue marble is drawn**
bet 1	free trip to WDC	free trip to LA
bet 2	free trip to NOLA	free trip to NYC

Let's also say that this decision maker has gone through the process to determine the point at which he or she is indifferent between the trip to New Orleans and a lottery between Washington, DC and Los Angeles, and the resulting utility value for New Orleans is .35. So, this person has these utility values:

a trip to Washington, DC	1
a trip to New York City	.75
a trip to New Orleans	.35
a trip to Los Angeles	0

If we replace the outcome with the value of the outcome in the decision table, we have this:

	P = .40 **red marble is drawn**	P = .60 **blue marble is drawn**
bet 1	1	0
bet 2	.35	.75

We then use the same method as we do to calculate the expected value of each option, except with utility values instead of monetary values.

EU(bet 1) = (.40)(1) + (.60)(0) = .40 + 0 = .40

EU(bet 2) = (.40)(.35) + (.60)(.75) = .14 + .45 = **.59**

Following the expected utility strategy, the decision-maker picks the option with the highest expected utility, which is bet 2.

(Note that when we use the expected utility strategy, the expected utilities will always be between 0 and 1.)